Bowen Yang

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Research Interests

Robot learning for manipulation: vision–language–action models, demonstration collection, and the sensing, calibration, and control layers beneath learned policies.

Education

Shanghai Jiao Tong University Aug 2023 –
B.A. in French; B.Eng. in Information Engineering (minor degree) Jun 2027 (expected)
SJTU Paris Elite Institute of Technology. Advisor: **Prof. Junchi Yan**.
Average score: **91.69** · Rank: **2/77**
Selected coursework (score/100): Mathematical Foundation for Artificial Intelligence (99), Advanced Linear Algebra (97), Advanced Electronics (97), Probability and Statistics (96), Data Structure (96), Computer Architecture (96), Optimisation (94).

Publications

GuidedVLA: Specifying Task-Relevant Factors via Plug-and-Play Action Attention Specialization 2026
Xiaosong Jia*, **Bowen Yang***, Zuhao Ge*, Xian Nie*, Yuchen Zhou*, Cunxin Fan*, Yufeng Li, Yilin Chai, Chao Jing, Zijian Liang, Qingwen Bu, Haidong Cao, Chao Wu, Qifeng Li, Zhenjie Yang, Chenhe Zhang, Hongyang Li, Zuxuan Wu†, Junchi Yan†, Yu-Gang Jiang†
Robotics: Science and Systems (RSS)

ELAN4D: Embodiment-Centric 4D Supervision for Vision-Language-Action Models via Plug-and-Play Adaptation 2026
Zeyuan He*, **Bowen Yang***, Zhirui Fang*, Keru Zhou*, Lei Jiang, Jingjing Qian, Fan Mo, Junchi Yan, Philip Torr, Xiu Li, Li Jiang, Jialin Yu†
Conference on Robot Learning (CoRL)

FOMI: A Force-Feedback Manipulation Interface for Teaching Contact-Rich Manipulation 2026
Yishu Shen, Yiming Luo, Ruilin Wang, **Bowen Yang**, Shuai Yang, Fangyi Xu, Yujun Shen, Fu Zhang, Boyu Zhou, Yinghao Xu†
arXiv preprint arXiv:2610.19839
* Equal Contribution; † Corresponding Author

Research Experience

RobbyAnt (Ant Group), Algorithm Engineer Intern Jun 2026 – Present
Contact-rich manipulation: demonstration collection and policy learning · Co-author · FOMI, arXiv 2026

- Built **80 Hz synchronized** multi-sensor acquisition with a **data-check tool**, and extended **tracker-to-robot calibration** (forward kinematics, nonlinear least squares).
- Collected **robot-free and teleop demonstrations** and built the **PICO 4 Ultra VR baseline**.
- Implemented **relative-pose actions** and **real-time chunking**. Co-trained $\boldsymbol{\pi_{0.5}}$ **policies**.
- Deployed and evaluated policies on **Franka and Flexiv**: **78.4%** average success on four tasks (100 demonstrations each), and **68.2%** after transfer to a dual-arm Flexiv.

RethinkLab, Shanghai Jiao Tong University, Research Assistant Sep 2025 – May 2026
Advisors: Prof. Junchi Yan and Prof. Xiaosong Jia.

GuidedVLA, co-first author Sep 2025 – Jan 2026
Task-relevant representation learning for VLA policies · RSS 2026 · Project

- Led **codebase implementation and simulation experiments** for explicit supervision of task-relevant factors in a vision–language–action policy.

- Implemented **specialized action-attention heads** for **object grounding**, **temporal skill recognition**, and **depth-based geometric perception** in the π_0 action decoder.
- Integrated a **ControlNet-style residual adapter** for plug-and-play factor supervision.
- Evaluated against π_0: average success rose from **68.2% to 75.4%** on LIBERO-Plus and from **77.4% to 90.6%** on RoboTwin 2.0.

ELAN4D, co-first author — Mar 2026 – May 2026
4D supervision for VLAs · CoRL 2026 · with **Torr Vision Group, University of Oxford**

- Proposed **URDF-based forward kinematics** to turn proprioception into future **robot-joint and end-effector keypoint tracks** as low-cost **4D supervision**.
- Proposed a **ControlNet-style residual branch** that **isolates auxiliary gradients** from the pretrained VLM and keeps the base policy's inference interface unchanged.
- Co-designed **ablations** on 4D supervision, prediction location, and **robot-keypoint versus whole-scene tracking** to isolate the effect of each component.
- LIBERO-Plus success rose from **53.6% to 67.6%** (π_0) and **73.6% to 78.2%** ($\pi_{0.5}$).

Selected Projects

RoboMaster University Championship — Team Jiao Loong — Sep 2023 – Aug 2025
Vice-Captain & Algorithm Lead (Aug 2024 – Aug 2025)

- Led a **15-member algorithm team** that became **national champions in 2024 and 2025**. Also recruited and trained **60+ members** and ran a **32-team on-campus competition**.
- Developed **FAST-LIO2 LiDAR–inertial SLAM** with **GICP map registration** and **PTP-synchronized LiDAR timing** for localization during fast robot motion.
- Built a motion planner combining **kinodynamic A$*$** and **model predictive control (MPC)** for navigation and trajectory tracking on embedded hardware.
- Deployed **TensorRT YOLO** on **Jetson Orin**, with **EKF tracking** of occluded robots.

Honors & Awards

National Scholarship — 2024, 2025
Ministry of Education of China

National Champion, RoboMaster University Championship — 2024, 2025
Team Jiao Loong, Shanghai Jiao Tong University

RM Award (Algorithm), RoboMaster University Championship — 2025
Individual technical honor; one of three algorithm recipients in 2025

Finalist, Mathematical Contest in Modeling (MCM) — 2025

Bronze Medal, National Olympiad in Informatics (NOI) Winter Camp — 2021
China Computer Federation (CCF)

Platinum Division, USACO — 2020
USA Computing Olympiad (USACO)

Technical Skills

Programming: Python, C, C++
Machine Learning & Vision: PyTorch, OpenCV, TensorRT
Robotics & Development: ROS/ROS 2, Docker, Git, LaTeX
Robots & Hardware: Franka, Flexiv, Jetson Orin, LiDAR, PICO 4 Ultra
Simulation & Benchmarks: LIBERO, LIBERO-Plus, RoboTwin 2.0

Languages

Chinese: Native.
English: TOEFL iBT **5.5/6** (ETS comparable total: **110/120**), 8 Mar 2026.
French: DELF B2 (CEFR B2), diploma issued 26 Apr 2026.